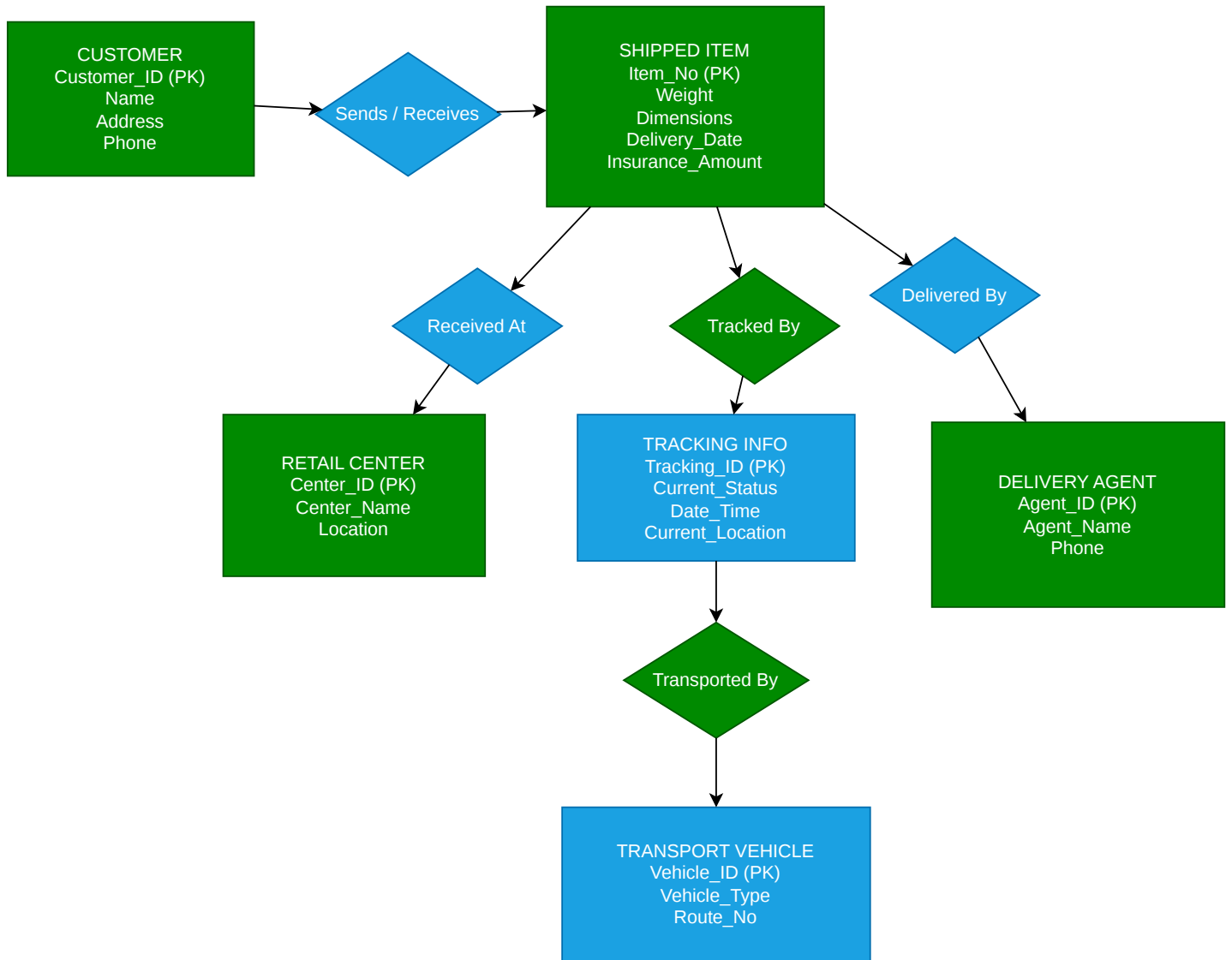
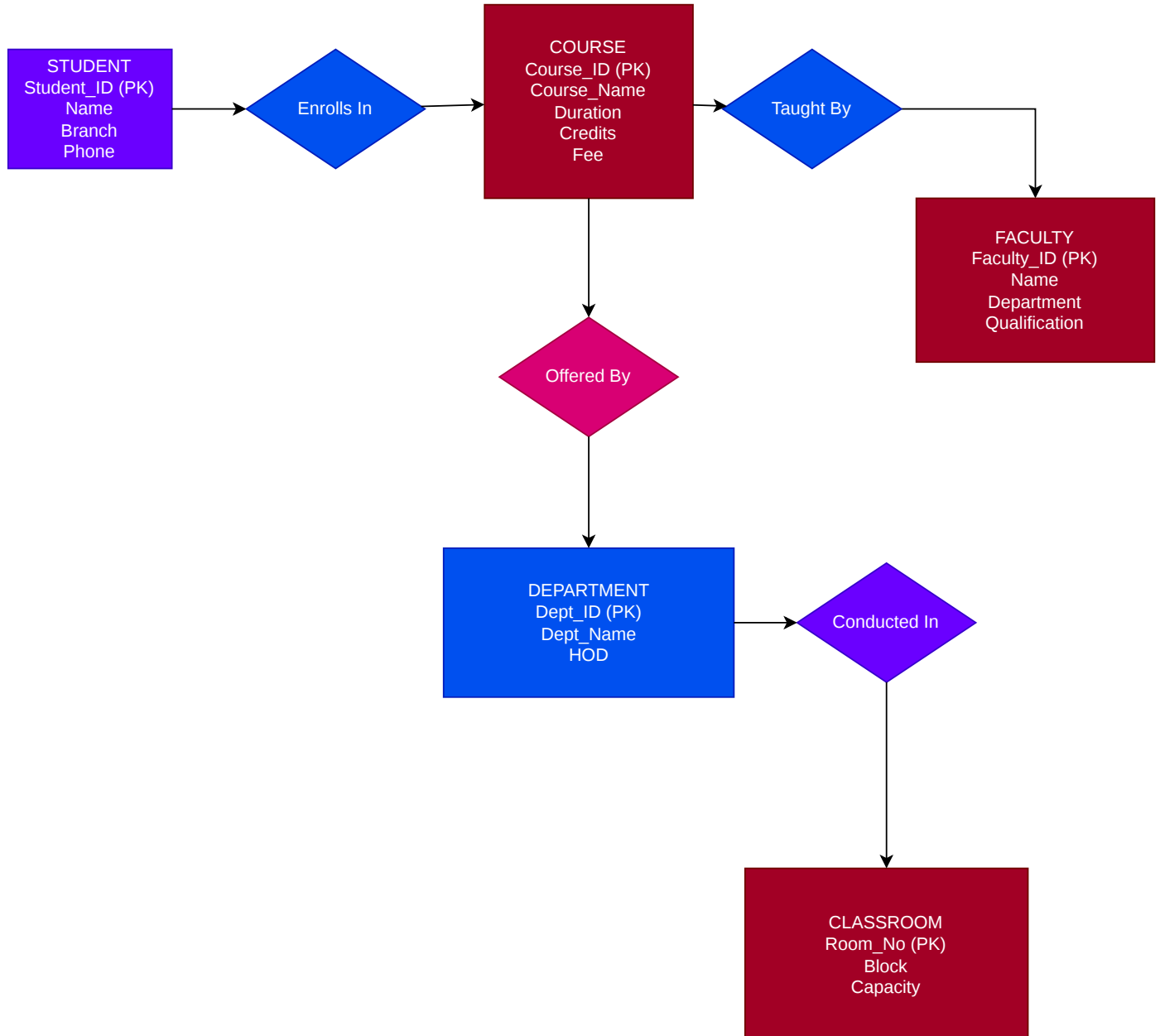


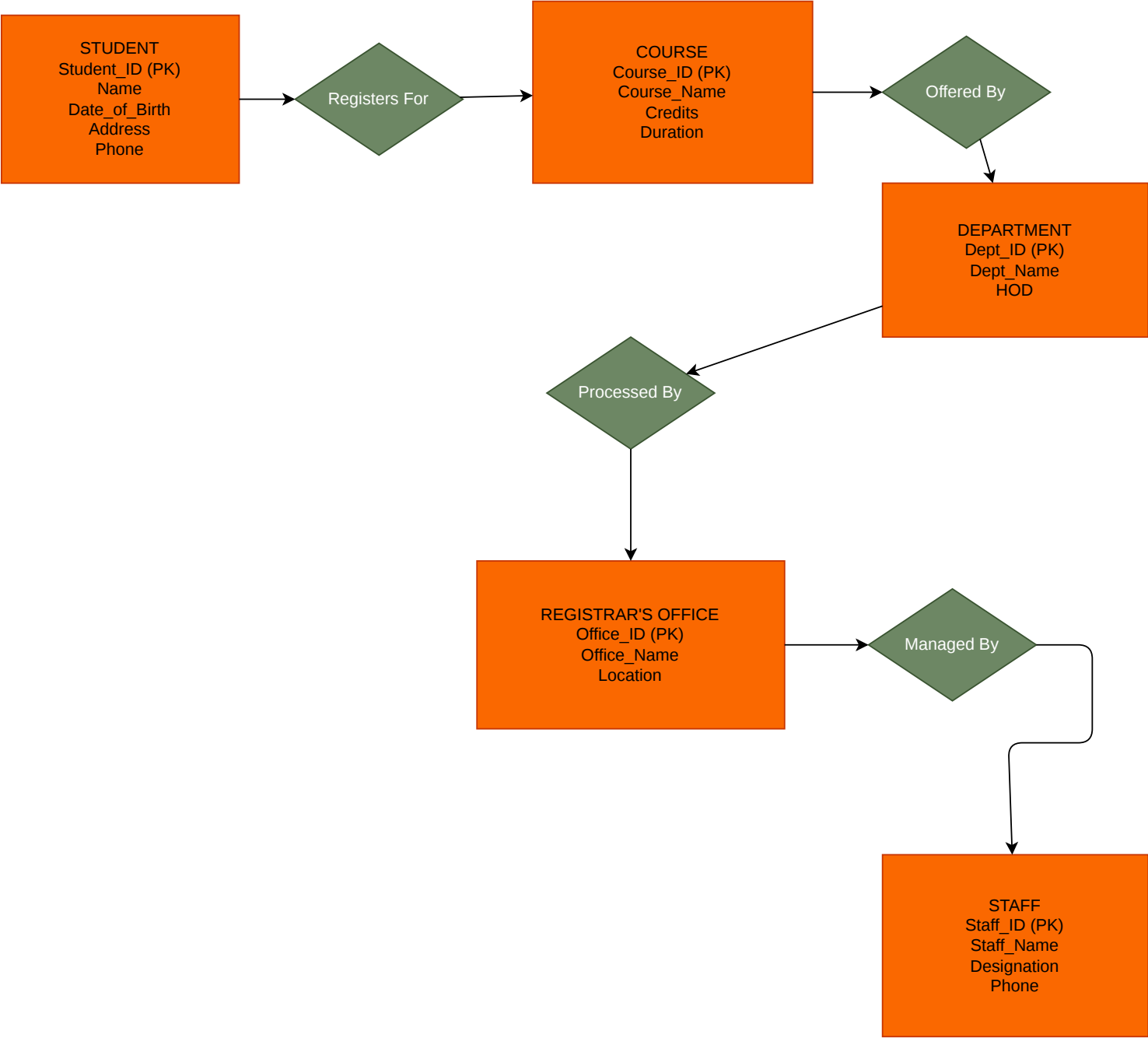
UPS Product Tracking Information System



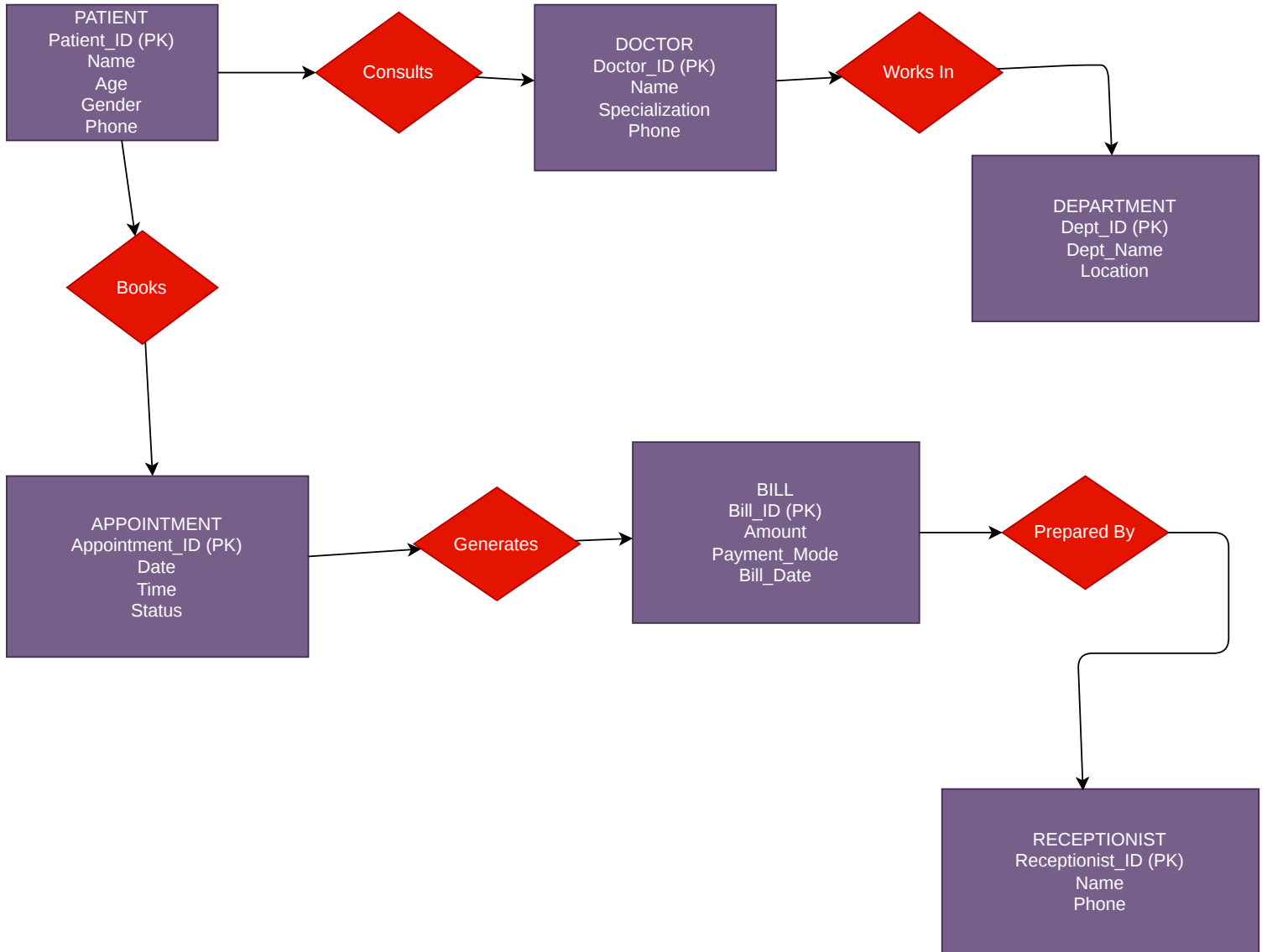
Course System Management



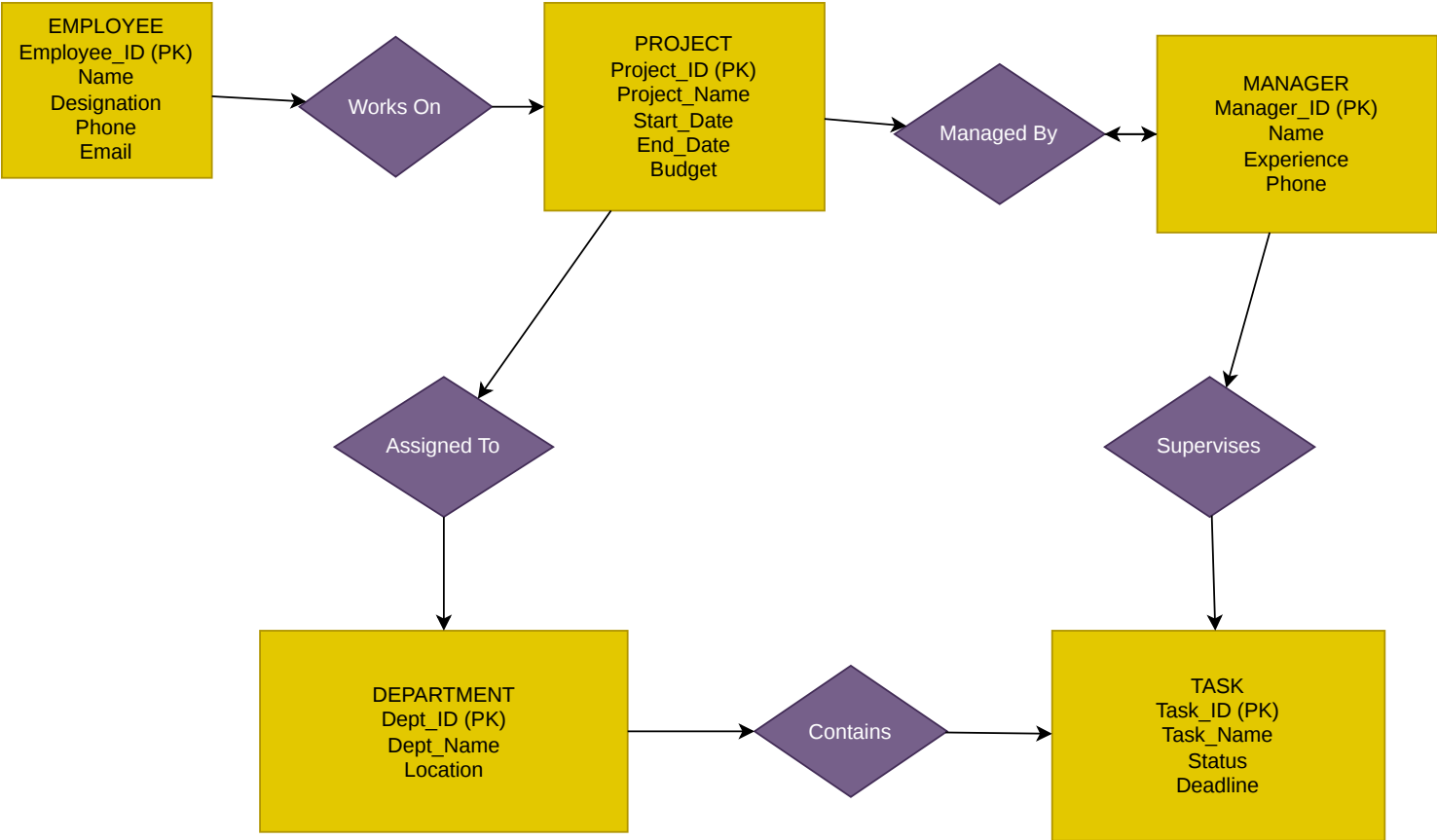
University Registrar's Office



Hospital Management System



Company Project Management System



6. Schema and Instance

Schema

A Schema is the blueprint or logical structure of a database. It defines the tables, attributes, relationships, and constraints of the database. A schema is designed before data is stored and changes very rarely.

Example Schema

STUDENT (Roll_No, Name, Department, Marks)

Instance

An Instance is the actual data stored in the database at a particular point in time. It changes whenever records are inserted, updated, or deleted.

Example

Roll_No	Name	Department	Marks
7	Chaithu	CSE	95
8	Kundhana	EEE	77

Types of Schema

1. Logical Schema

- * Describes the logical structure of the database.
- * Defines tables, attributes, relationships, and constraints.

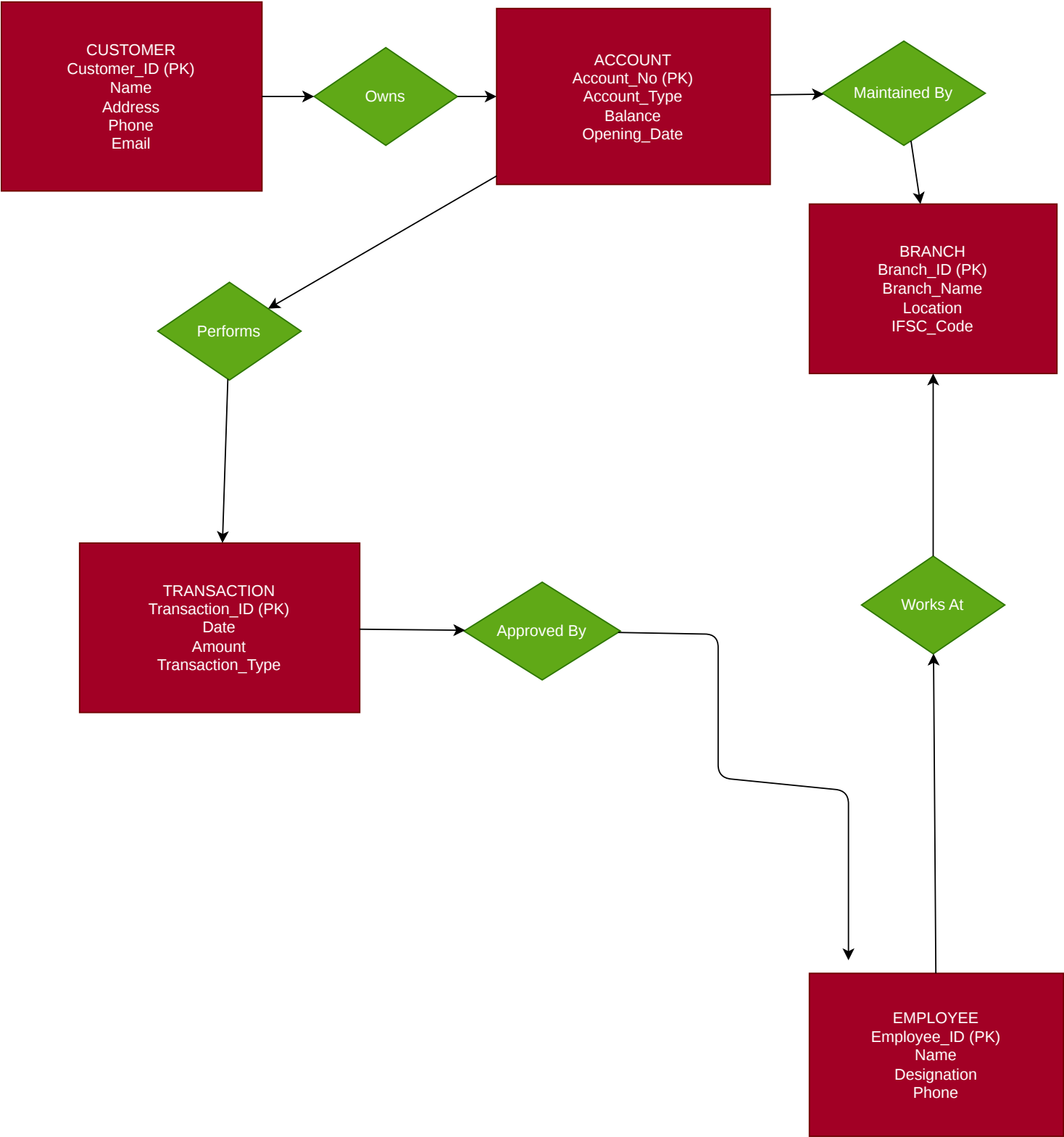
2. Physical Schema

- * Describes how data is physically stored on disk.
- * Includes files, indexes, storage blocks, and access methods.
- * Managed by the DBMS.

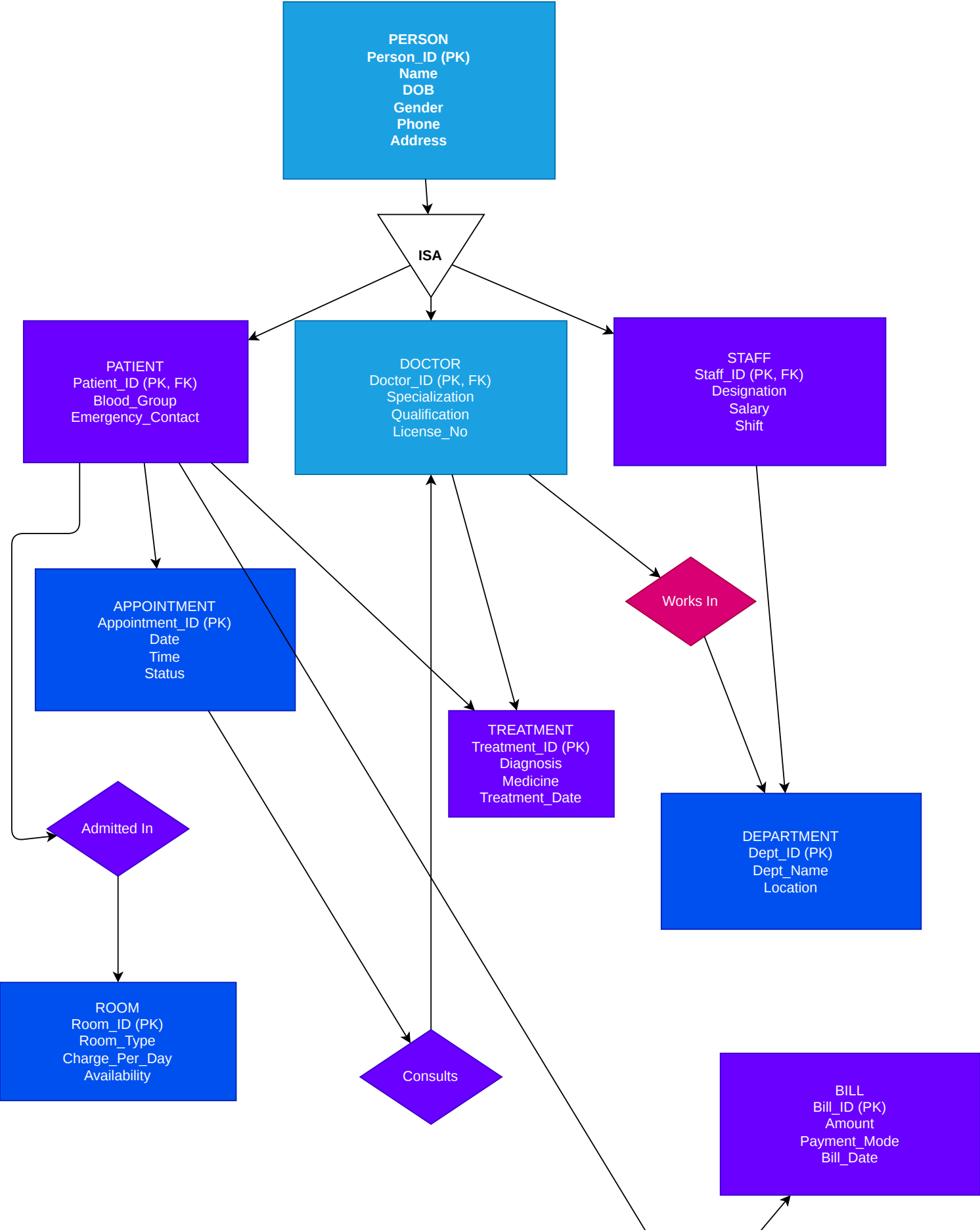
3. View Schema

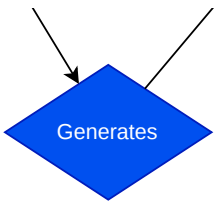
- * Defines what data different users can access.
- * Provides security by hiding unnecessary information.
- * Different users can have different views of the same database.

Bank Management System

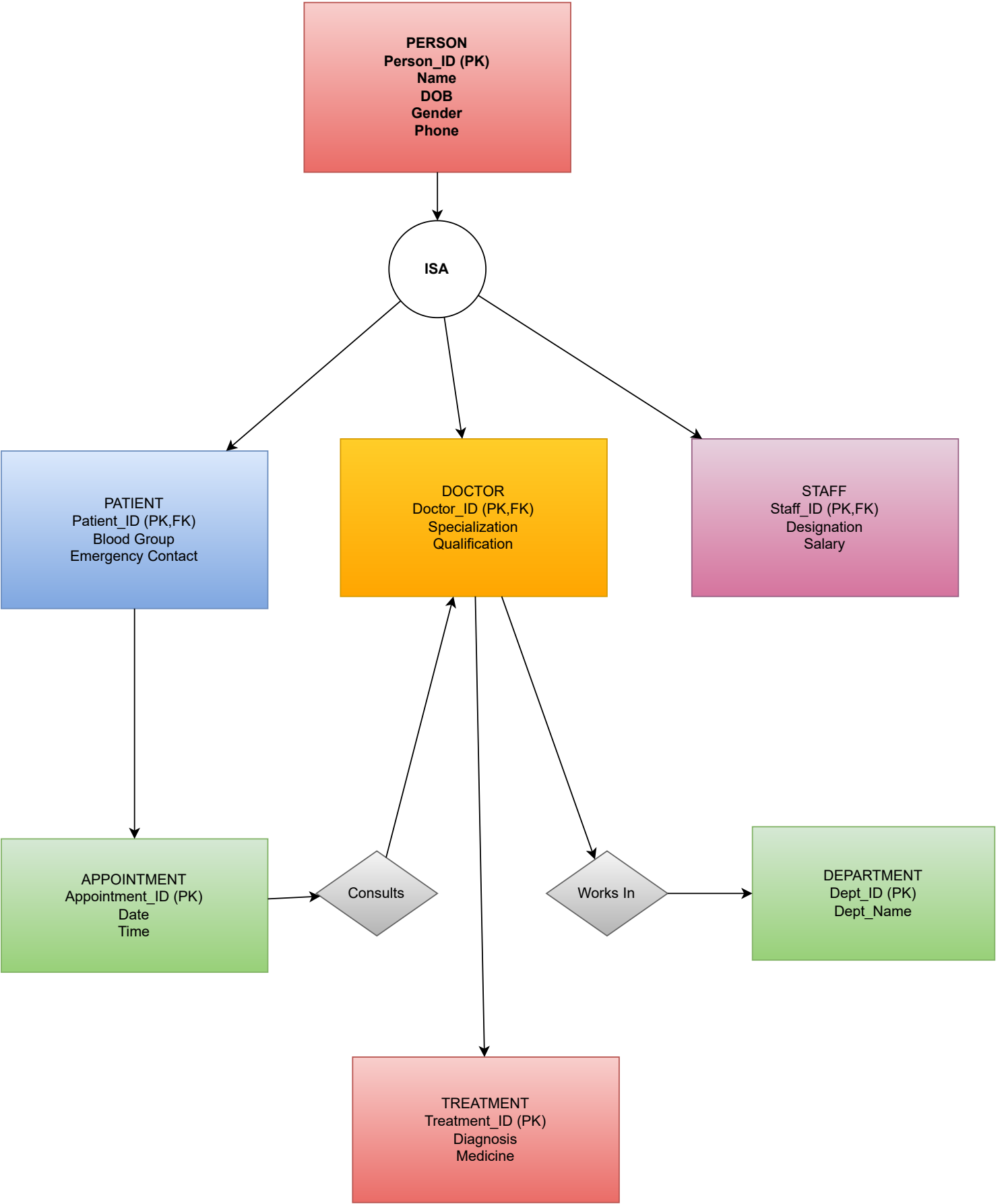


Hospital Management System (EER Diagram)





Hospital Management System (EER)



10. Role of DBA (Database Administrator)

Definition

A Database Administrator (DBA) is responsible for managing, securing, maintaining, and controlling the database to ensure its efficient, reliable, and secure operation.

Responsibilities of DBA

- * Install and configure the Database Management System (DBMS).
- * Create, modify, and maintain databases.
- * Manage user accounts, roles, and access permissions.
- * Perform database backup and recovery.
- * Ensure database security and prevent unauthorized access.
- * Monitor and improve database performance.